

Compiler Design Lab Programs

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Course Code: CSA1402

28.In a class, an English teacher was teaching the vowels and consonants to the students. She says “Vowel sounds allow the air to flow freely, causing the chin to drop noticeably, whilst consonant sounds are produced by restricting the air flow”. As a class activity the students are asked to identify the vowels and consonants in the given word/sentence and count the number of elements in each. Write an algorithm to help the student to count the number of vowels and consonants in the given sentence.

Code:

```
%{  
#include <stdio.h>  
  
int vowels=0, consonants=0;  
  
%}  
%%  
[AEIOUaeiou] { vowels++; }  
[a-zA-Z]     { consonants++; }  
.\n        ;  
%%  
  
int main(){ yylex(); printf("Vowels: %d\nConsonants: %d\n", vowels, consonants); return 0; }  
int yywrap(){ return 1; }
```

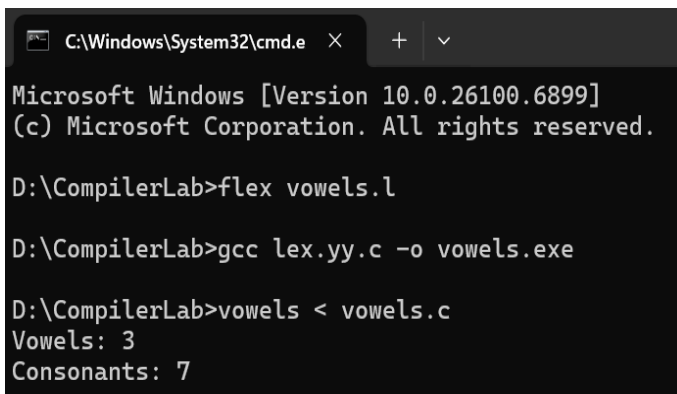
Output:

Hello World

Vowels: 3

Consonants: 7

Output Screenshot:



```
C:\Windows\System32\cmd.e X + v  
Microsoft Windows [Version 10.0.26100.6899]  
(c) Microsoft Corporation. All rights reserved.  
  
D:\CompilerLab>flex vowels.l  
  
D:\CompilerLab>gcc lex.yy.c -o vowels.exe  
  
D:\CompilerLab>vowels < vowels.c  
Vowels: 3  
Consonants: 7
```

29. Keywords are predefined, reserved words used in programming that have special meanings to the compiler. Keywords are part of the syntax and they cannot be used as an identifier. In general there are 32 keywords. The prime function of Lexical Analyser is token

Generation. Among the 6 types of tokens, differentiating Keyword and Identifier is a challenging issue. Thus write a LEX program to separate keywords and identifiers.

Code:

```
%{  
#include <stdio.h>  
%}  
  
KEYWORD  
(int|float|char|double|void|return|if|else|for|while|break|continue|switch|case|printf|scanf|main)  
  
ID    [a-zA-Z_][a-zA-Z0-9_]*  
  
%%  
  
{KEYWORD}  { printf("Keyword: %s\n", yytext); }  
{ID}       { printf("Identifier: %s\n", yytext); }  
  
[ \t\n;, \(\)\{\}\] ;  
  
.  
%%  
  
int main(){ yylex(); return 0; }  
  
int yywrap(){ return 1; }
```

Output:

```
int number; float avg; printf("ok");
```

```
Keyword: int
```

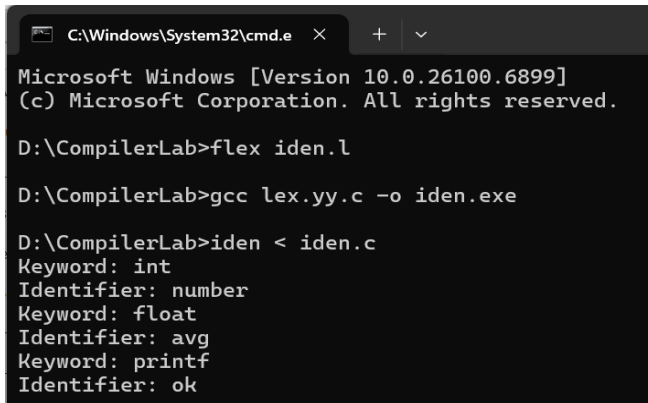
```
Identifier: number
```

```
Keyword: float
```

```
Identifier: avg
```

```
Keyword: printf
```

Output Screenshot:



```
C:\Windows\System32\cmd.e
Microsoft Windows [Version 10.0.26100.6899]
(c) Microsoft Corporation. All rights reserved.

D:\CompilerLab>flex iden.l

D:\CompilerLab>gcc lex.yy.c -o iden.exe

D:\CompilerLab>iden < iden.c
Keyword: int
Identifier: number
Keyword: float
Identifier: avg
Keyword: printf
Identifier: ok
```

30. Write a LEX program to recognise numbers and words in a statement. Pooja is a small girl of age 3 always fond of games. Due to the pandemic, she was not allowed to play outside. So her mother designs a gaming event by showing a flash card. Pooja has to separate the numbers in one list and words in another list shown in the flash card.

Code:

```
%{
#include <stdio.h>
%}
%%

[0-9]+ { printf("Number: %s\n", yytext); }
[a-zA-Z]+ { printf("Word: %s\n", yytext); }
.\n    ;
%%

int main() { yylex(); return 0; }
int yywrap() { return 1; }
```

Output:

Pooja is 3 years old and has 5 toys

Copy code

Word: Pooja

Word: is

Number: 3

Word: years

Word: old

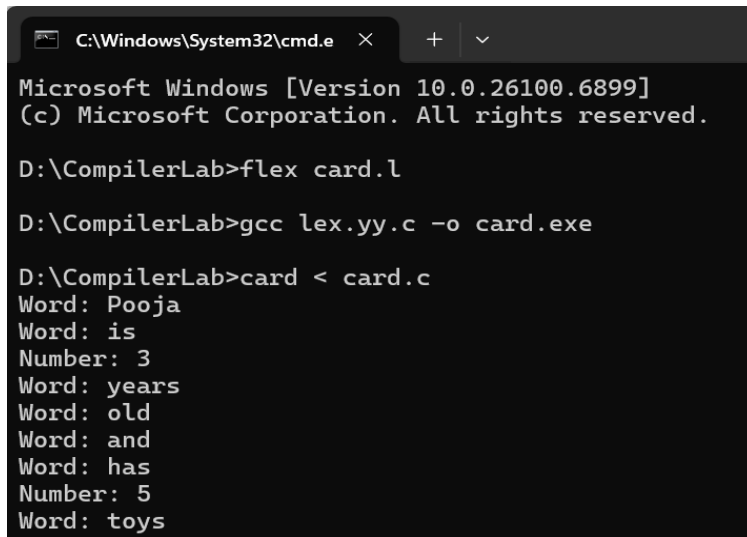
Word: and

Word: has

Number: 5

Word: toys

Output Screenshot:



```
C:\Windows\System32\cmd.e
Microsoft Windows [Version 10.0.26100.6899]
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D:\CompilerLab>flex card.l

D:\CompilerLab>gcc lex.yy.c -o card.exe

D:\CompilerLab>card < card.c
Word: Pooja
Word: is
Number: 3
Word: years
Word: old
Word: and
Word: has
Number: 5
Word: toys
```

31. Write a LEX program to identify and count positive and negative numbers.

Code:

```
%{
#include <stdio.h>

int pos=0, neg=0;
%}

%%

-[0-9]+ { neg++; printf("Negative: %s\n", yytext); }
[0-9]+ { pos++; printf("Positive: %s\n", yytext); }
.\n ;
%%

int main(){ yylex(); printf("Total Positive: %d\nTotal Negative: %d\n", pos, neg); return 0; }

int yywrap(){ return 1; }
```

Output:

-5 10 0 -8 25

Negative: -5

Positive: 10

Positive: 0

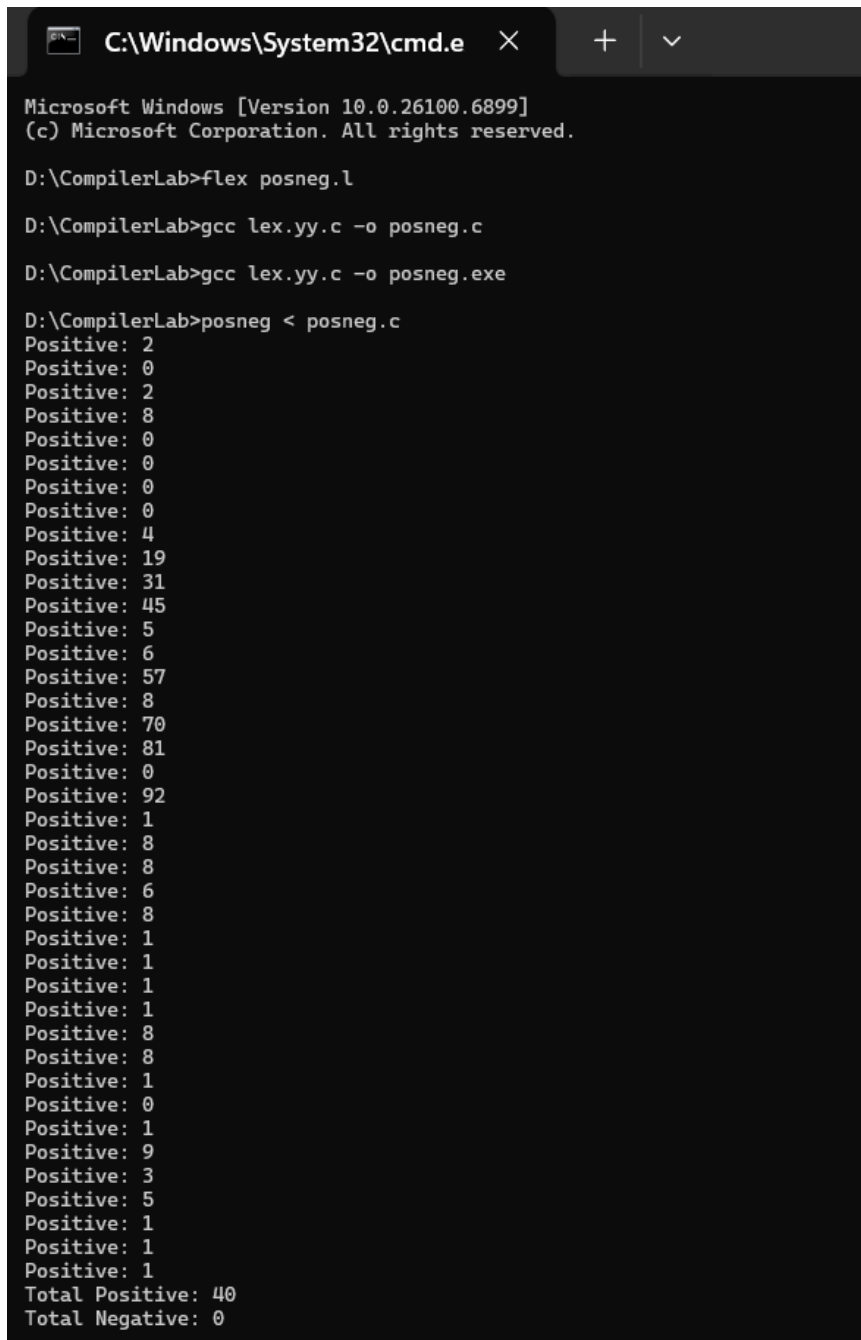
Negative: -8

Positive: 25

Total Positive: 3

Total Negative: 2

Output Screenshot:



```
C:\Windows\System32\cmd.e X + v
Microsoft Windows [Version 10.0.26100.6899]
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D:\CompilerLab>flex posneg.l

D:\CompilerLab>gcc lex.yy.c -o posneg.c

D:\CompilerLab>gcc lex.yy.c -o posneg.exe

D:\CompilerLab>posneg < posneg.c
Positive: 2
Positive: 0
Positive: 2
Positive: 8
Positive: 0
Positive: 0
Positive: 0
Positive: 0
Positive: 4
Positive: 19
Positive: 31
Positive: 45
Positive: 5
Positive: 6
Positive: 57
Positive: 8
Positive: 70
Positive: 81
Positive: 0
Positive: 92
Positive: 1
Positive: 8
Positive: 8
Positive: 6
Positive: 8
Positive: 1
Positive: 1
Positive: 1
Positive: 1
Positive: 8
Positive: 8
Positive: 1
Positive: 0
Positive: 1
Positive: 9
Positive: 3
Positive: 5
Positive: 1
Positive: 1
Positive: 1
Total Positive: 40
Total Negative: 0
```

32.A networking company wants to validate the URL for their clients. Write a LEX program to implement the same.

Code:

```
%{
#include <stdio.h>

%}

%%

(https?://)?(www\.)?[a-zA-Z0-9.-]+\.[a-zA-Z]{2,}(/[a-zA-Z0-9._-]*)* { printf("Valid URL:
%s\n", yytext); }

[^\n]+ { printf("Invalid URL: %s\n", yytext); }

\n ;

%%

int main(){ yylex(); return 0; }

int yywrap(){ return 1; }
```

Output:

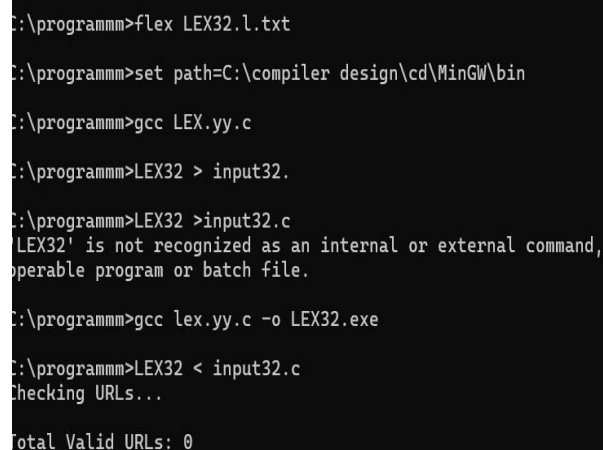
https://www.example.com/path

[ftp://bad.url](#)

Valid URL: https://www.example.com/path

Invalid URL: ftp://bad.url

Output Screenshot:



```
C:\programmm>flex LEX32.l.txt
C:\programmm>set path=C:\compiler design\cd\MinGW\bin
C:\programmm>gcc LEX.yy.c
C:\programmm>LEX32 > input32.c
C:\programmm>LEX32 >input32.c
'LEX32' is not recognized as an internal or external command,
operable program or batch file.
C:\programmm>gcc lex.yy.c -o LEX32.exe
C:\programmm>LEX32 < input32.c
Checking URLs...
Total Valid URLs: 0
```